✅ **1. Competency #6: Identify and describe the mathematical aspects of a real-world problem situation, including identifying the significant variables**

Highly relevant (core competency)

- Students must extract multiple key mathematical elements from the text:
  - Distance: uphill = 9 km, downhill = 9 km (total round trip = 18 km)
  - Speeds: uphill = 1.5 km/h, downhill = 3 km/h
  - Time constraint: must return by 8:00 PM
  - Target quantity: latest departure time
- These are "significant variables" embedded in narrative language that must be recognized and given mathematical meaning.

✅ **2. Competency #2: Identify the key variables in a model**

Highly relevant

- Clear variables in the model include:
  - $d_{\text{up}} = 9$, $v_{\text{up}} = 1.5 \rightarrow t_{\text{up}} = d/v = 6$ hours
  - $d_{\text{down}} = 9$, $v_{\text{down}} = 3 \rightarrow t_{\text{down}} = 3$ hours
  - Total time $T = t_{\text{up}} + t_{\text{down}} = 9$ hours
  - Departure time $= 20:00 - 9\,\text{h} = 11:00\ \text{AM}$
- The variables are linked through functional relationships (time = distance ÷ speed), which is foundational to modeling.

✅ **3. Competency #7: Simplify or decompose a situation or problem to make it amenable to mathematical analysis**

Highly relevant

- The problem is effectively decomposed into two distinct phases: ascent and descent.
- Although real-world walking speeds may vary, the problem states that the given speeds already include rest and meal breaks. Thus, students can simplify the entire journey into a piecewise constant-speed model.
- This "divide-and-conquer" approach is a key strategy in mathematical modeling.

✅ 4. Competency #12: Identify the constraints, assumptions, and simplifications in a mathematical model

Moderately to highly relevant

- Implicit assumptions include:
  - Uphill and downhill distances are equal (9 km each)
  - Speeds are constant and already incorporate rest periods (no extra time needed)
  - The hiker must complete the entire round trip before 8:00 PM (a hard deadline)
- Constraints are explicit: total duration ≤ 9 hours; finish time ≤ 20:00.
- While the question doesn't ask students to state these explicitly, recognizing them is essential for correct modeling.

✅ 5. Competency #9: Translate a problem into a standard mathematical representation or algorithm

Moderately relevant

- Students must convert the verbal description into the standard physics formula:
  $t = \frac{d}{v}$ (time = distance ÷ speed)
- Then assemble a multi-step algorithm:
  1. Compute ascent time
  2. Compute descent time
  3. Sum to get total time
  4. Subtract total time from 20:00 → obtain latest start time
- This is a clear process of translating a real-world scenario into a computational procedure.

✅ 6. Competency #1: Select a mathematical description or representation that describes a problem

Moderately relevant

- Students must recognize this as a distance-rate-time (DRT) problem, not a proportional reasoning, statistical, or geometric one.
- Choosing a "piecewise uniform motion model" as the descriptive framework is a meaningful modeling decision.

✅ 1. Competency #6: Identify and describe the mathematical aspects of a real-world problem situation, including identifying the significant variables

高度相关（核心）

- 学生必须从文本中识别出多个关键数学要素：
  - 路程：上山 = 9 km，下山 = 9 km（总 18 km）
  - 速度：上山 = 1.5 km/h，下山 = 3 km/h
  - 时间约束：必须在 8:00 PM 前返回
  - 所求量：最晚出发时间
- 这些都是隐藏在叙述中的“显著变量”，需主动提取并赋予数学意义。

✅ 2. Competency #2: Identify the key variables in a model

高度相关

- 明确模型中的变量：
  - $d_{\text{up}} = 9$, $v_{\text{up}} = 1.5 \rightarrow t_{\text{up}} = d/v = 6$ 小时
  - $d_{\text{down}} = 9$, $v_{\text{down}} = 3 \rightarrow t_{\text{down}} = 3$ 小时
  - 总耗时 $T = t_{\text{up}} + t_{\text{down}} = 9$ 小时
  - 出发时间 $= 20:00 - 9\,\text{h} = 11:00\ \text{AM}$
- 变量之间存在明确的函数关系（时间 = 距离 ÷ 速度），这是建模的基础。

✅ 3. Competency #7: Simplify or decompose a situation or problem to make it amenable to mathematical analysis

高度相关

- 问题被有效分解为两个独立阶段：上山 + 下山。
- 尽管现实中速度可能波动，但题目已说明“平均速度包含休息”，因此学生可将整个过程简化为匀速分段模型。
- 这种“分而治之”的策略是数学建模的关键技巧。

✅ 4. Competency #12: Identify the constraints, assumptions, simplifications in a mathematical model

中高相关

- 隐含假设包括：
  - 上/下山路径长度相等（9 km each）
  - 速度恒定且已包含休息（无需额外加休息时间）
  - 必须完整往返 before 8 pm（硬性截止时间）
- 约束条件明确：总行程 ≤ 9 小时，结束时间 ≤ 20:00。
- 虽然题目未要求学生写出这些假设，但理解它们对正确建模至关重要。

✅ 5. Competency #9: Translate a problem into a standard mathematical representation or algorithm

中等相关

- 学生需将文字描述转化为标准物理公式：
  $t = \frac{d}{v}$ （时间 = 路程 ÷ 速度）
- 然后组合为算法：
  1. 计算上山时间
  2. 计算下山时间
  3. 求和得总时间
  4. 从 20:00 反向减去总时间 → 得出发时间
- 这是一个清晰的多步算法转化过程。

✅ 6. Competency #1: Select a mathematical description or representation that describes a problem

中等相关

- 学生需判断：此问题属于“行程问题”（distance-rate-time），而非比例、统计或几何。
- 选择“分段匀速运动模型”作为描述框架，是建模的重要决策。